Ericsson PE Node BAS-PE-10 Basavanagudi POP Routing Protocol Flap SOP

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_9860

Description

This SOP addresses a Critical Failure Domain involving Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Ericsson PE Node BAS-PE-10 at Basavanagudi POP. The underlying cause of this failure domain is attributed to the inherent complexities of dynamic routing protocols and the potential for flapping or oscillations in the control plane, which can lead to instability and suboptimal routing. In the context of Ericsson's CLI dialect, this failure domain is often precipitated by the interaction between BGP's Route Refresh and the OSPF/ISIS LSP flooding mechanisms. The PE Node's role in facilitating inter-domain routing and its reliance on these protocols make it particularly susceptible to this type of failure. Furthermore, the specific equipment and configuration at Basavanagudi POP may introduce additional layers of complexity, necessitating a nuanced and vendor-specific approach to diagnosis and resolution.

Pre-requisite Checks

- check: Verify the status of redundant links and ensure they are operational and not contributing to the failure., action: Dispatch remote hands to Basavanagudi POP to perform physical checks.

- check: Confirm that the Ericsson PE Node BAS-PE-10 is properly powered and its hardware is within specified temperature and humidity ranges., action: Review the site's environmental monitoring systems and ensure that all equipment is within normal operating parameters.

Potential Root Causes

Cause	Probability
BGP Route Refresh leading to OSPF/ISIS LSP flooding and control plane instability.	High
Misconfigured or malfunctioning BGP peers leading to flapping or oscillations in the control plane.	Medium
Hardware or software issues on the Ericsson PE Node BAS-PE-10 or adjacent devices.	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp neighbor, Expected Output: Neighbor AS RouterID
State UP-DOWN LocalAS MsgRcvdMsgSentMsgs OutQ InQ, Description: This
command is used to verify the status of BGP neighbors and identify any
potential issues that may be contributing to the failure domain. The
output should indicate the state of each neighbor, including any
errors or warnings that may be relevant to the diagnosis.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command is used to
verify the status of OSPF neighbors and identify any potential issues
that may be contributing to the failure domain. The output should
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```
indicate the state of each neighbor, including any errors or warnings
that may be relevant to the diagnosis.

Command: show isis neighbor, Expected Output: Neighbor IP address
State Dead time Interface, Description: This command is used to verify
the status of ISIS neighbors and identify any potential issues that
may be contributing to the failure domain. The output should indicate
the state of each neighbor, including any errors or warnings that may
be relevant to the diagnosis.
```

Resolution Steps

Step 1: Step: Verify and Correct BGP Neighbor Configuration, Command: neighbor remote-as , Description: This step involves verifying the BGP neighbor configuration and correcting any errors or misconfigurations that may be contributing to the failure domain. The command will update the BGP neighbor configuration to ensure that it is correctly set up and functioning as expected.

Step 2: Step: Disable OSPF/ISIS LSP Flooding, Command: ospf flooding disable, Description: This step involves disabling the OSPF/ISIS LSP flooding mechanism to prevent further instability in the control plane. The command will disable the LSP flooding feature and prevent it from contributing to the failure domain.

Step 3: Step: Restart the Ericsson PE Node BAS-PE-10, Command: restart node , Description: This step involves restarting the Ericsson PE Node BAS-PE-10 to ensure that it is properly rebooted and functioning as expected. The command will restart the node and allow it to come up with a clean configuration and state.

Post-Check Verification

```
Command: show bgp neighbor, Expected Output: Neighbor AS RouterID
State UP-DOWN LocalAS MsgRcvdMsgSentMsgs OutQ InQ, Description: This
command is used to verify that the BGP neighbor configuration has been
corrected and that the neighbors are functioning as expected.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command is used to
verify that the OSPF neighbor configuration has been updated and that
the neighbors are functioning as expected.

Command: show isis neighbor, Expected Output: Neighbor IP address
State Dead time Interface, Description: This command is used to verify
that the ISIS neighbor configuration has been updated and that the
neighbors are functioning as expected.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: load running-config , Description: This step involves restoring the original configuration to the Ericsson PE Node BAS-PE-10 in the event that the resolution steps fail to resolve the issue. The command will load the original configuration from a file and restore the node to its previous state.

- Step: Reboot the Ericsson PE Node BAS-PE-10, Command: restart node , Description: This step involves rebooting the Ericsson PE Node BAS-PE-10 to ensure that it is properly restarted and functioning as expected in the event that the resolution steps fail to resolve the issue.